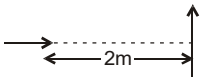
SYLLABUS : ELECTROMAGNETIC FORCE

1. Two identical short magnetic dipoles of magnetic moments 1.0 A-m^2 each, placed at a separation of 2 m with their axes perpendicular to each other. The resultant magnetic field at a point midway between the dipole is:



(A) 5×10^{-7} T (B) $\sqrt{5} \times 10^{-7}$ T (C) 10^{-7} T (D) 2×10^{-7} T

2. A point charge is moving in a circle with constant speed. Consider the magnetic field produced by the charge at a fixed point P (not centre of the circle) on the axis of the circle.

(A) it is constant in magnitude only

(B) it is constant in direction only

(C) it is constant in direction and magnitude both

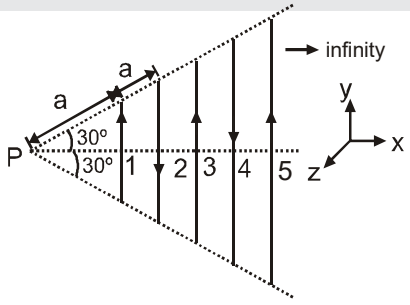
(D) it is not constant in magnitude and direction both.

3. The magnetic moment of a short dipole is, 1 A m^2. What is the magnitude of the magnetic induction in air at 10 cm from centre of the dipole on a line making an angle of 30º from the axis of the dipole?

(A) $\frac{\sqrt{7}}{2} \times 10^{-4}$ wb/m^2 (B) $\frac{\sqrt{13}}{3} \times 10^{-4}$ wb/m^2 (C) $\frac{\sqrt{13}}{2} \times 10^{-4}$ wb/m^2 (D) $\frac{\sqrt{15}}{2} \times 10^{-4}$ wb/m^2

4. Infinite number of straight wires each carrying current I are equally placed as shown in the figure. Adjacent wires have current in opposite direction. Net magnetic field at point P is

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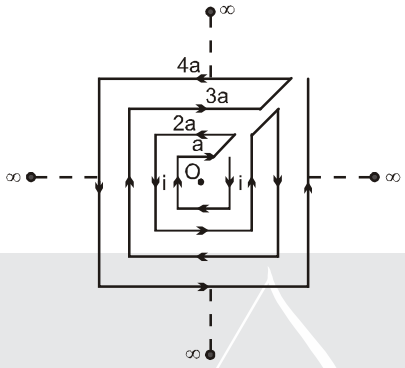


(A) $\frac{\mu_0 I}{4\pi} \frac{\ell n 2}{\sqrt{3}a} \hat{k}$ (B) $\frac{\mu_0 I}{4\pi} \frac{\ell n 4}{\sqrt{3}a} \hat{k}$ (C) $\frac{\mu_0 I}{4\pi} \frac{\ell n 4}{\sqrt{3}a} (-\hat{k})$ (D) zero

5. Two infinitely long, thin, insulated, straight wires lie in the x-y plane along the x and y-axis respectively. Each wire carries a current i, respectively in the positive x-direction and positive y-direction. The magnetic field will be zero at all points on the straight line:

(A) y = x (B) y = – x (C) y = x – 1 (D) y = – x + 1

6. Determine the magnitude of magnetic field at the centre of the current carrying wire arrangement shown in the figure. The arrangement extends to infinity. (The wires joining the successive squares are along the line passing through the centre)

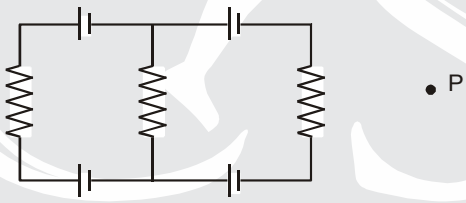


(A) $\frac{\mu_0 \, i}{\sqrt{2}\,\pi\, a}$ (B) 0 (C) $\frac{2\sqrt{2}\,\mu_0 \, i}{\pi\, a}\ln 2$ (D) none of these

7. A current of 1 amp is flowing in the sides of an equilateral triangle of side 4.5×10^{-2} m. Find the magnetic field at the centroid of triangle.

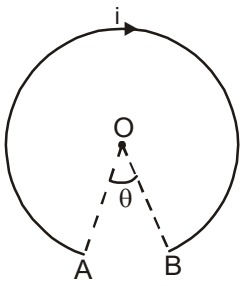
(A) 2×10^{-5} wb/m^2 (B) 3×10^{-5} wb/m^2 (C) 8×10^{-5} wb/m^2 (D) 4×10^{-5} wb/m^2

8. Each of the batteries shown in figure has an emf equal to 10 V. Find the magnetic field B at the point p.



(A) 0 (B) 1 (C) 2 (D) 5

9. A current carrying wire AB of the length $2\pi R$ is turned along a circle, as shown in figure. The magnetic field at the centre O.



(A) $\frac{\mu_0 i}{2R}\left(\frac{2\pi-\theta}{2\pi}\right)^2$ (B) $\frac{\mu_0 i}{2R}\left(\frac{2\pi-\theta}{2\pi}\right)$

(C) $\frac{\mu_0 i}{2R}(2\pi-\theta)$ (D) $\frac{\mu_0 i}{2R}(2\pi+\theta)^2$

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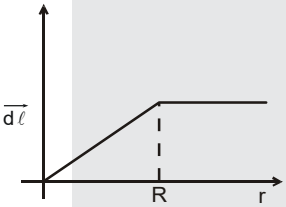
10. A wire is wound on a long rod of material of relative permeability μ_r = 4000 to make a solenoid. If the current through the wire is 5 A and number of turns per unit length is 1000 per metre, then the magnetic field inside the solenoid is :

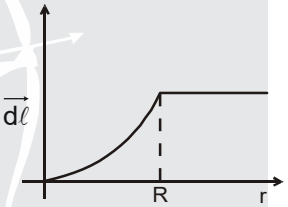
(A) 25.12 mT (B) 12.56 m T (C) 12.56 T (D) 25.12 T

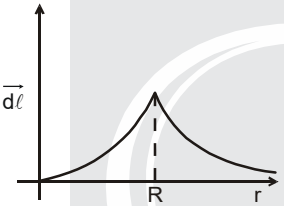
11. Axis of a solid cylinder of infinite length and radius R lies along y-axis it carries a uniformly distributed current ' i ' along +y direction. Magnetic field at a point $\left(\frac{R}{2}, y, \frac{R}{2}\right)$ is :-

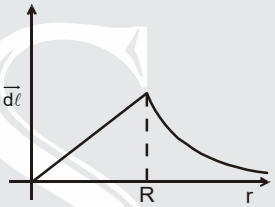
(A) $\frac{\mu_0 i}{4\pi R}(\hat{i}-\hat{k})$ (B) $\frac{\mu_0 i}{2\pi R}(\hat{j}-\hat{k})$ (C) $\frac{\mu_0 i}{4\pi R}\hat{j}$ (D) $\frac{\mu_0 i}{4\pi R}(\hat{i}+\hat{k})$

12. A cylindrical wire of radius R is carrying current i uniformly distributed over its cross-section. If a circular loop of radius ' r ' is taken as amperian loop, then the variation value of $\oint \vec{B}\cdot\vec{d\ell}$ over this loop with radius ' r ' of loop will be best represented by:

(A) 

(B) 

(C) 

(D) 

13. A constant direct current of uniform density $\vec{j}$ is flowing in an infinitely long cylindrical conductor. The conductor contains an infinitely long cylindrical cavity whose axis is parallel to that of the conductor and is at a distance $\vec{\ell}$ from it. What will be the magnetic induction $\vec{B}$ at a point inside the cavity at a distance $\vec{r}$ from the centre of cavity?

(A) $\frac{\mu_0\left(\vec{j}\times\vec{r}\right)}{2}$ (B) $\frac{\mu_0\left(\vec{j}\times\vec{\ell}\right)}{2}$

(C) $\frac{\mu_0\left(\vec{j}\times\vec{\ell}\right)+\mu_0\left(\vec{j}\times\vec{r}\right)}{2}$ (D) $\frac{\mu_0\left(\vec{j}\times\vec{\ell}\right)-\mu_0\left(\vec{j}\times\vec{r}\right)}{2}$

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14. A uniform current I flows along the length of an infinitely long, straight, thin walled pipe. Then

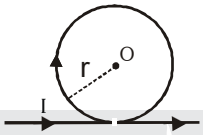
(A) the magnetic field at all points inside the pipe is the same, but not zero

(B) the magnetic field at any point inside the pipe is zero

(C) the magnetic field is zero only on the axis of the pipe

(D) the magnetic field is different at different points inside the pipe.

15. An infinitely long straight conductor is bent into the shape as shown in figure. It carries a current I ampere and the radius of the circular loop is r meter. Then the magnetic induction at the centre of the circular part is :–



(A) Zero (B) ∞ (C) $\frac{\mu_0}{4\pi}\frac{2I}{r}(\pi+1)$ (D) $\frac{\mu_0}{4\pi}\frac{2I}{r}(\pi-1)$

16. Electric current i enters and leaves a square loop made of homogeneous wire of uniform cross-section through diagonally opposite corners. A charge particle q moving along the axis of the square loop. Passes through centre at speed υ. The magnetic force acting on the particle when it passes through the centre has a magnitude

(A) $q\nu\frac{\mu_0 i}{2a}$ (B) $q\nu\frac{\mu_0 i}{2\pi a}$ (C) $q\nu\frac{\mu_0 i}{a}$ (D) zero

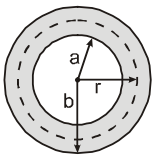
17. A negative charged particle falling freely under gravity enters a region having uniform horizontal magnetic field pointing towards north. The particle will be deflected towards

(A) East (B) West (C) North (D) South

18. A positively charged particle is moving along the positive X-axis. You want to apply a magnetic field for a short time so that the particle may reverse its direction and move parallel to the negative X-axis. This can be done by applying the magnetic field along.

(A) Y-axis (B) Z-axis (C) Y-axis only (D) Z-axis only.

19. Figure shows a cylindrical conductor of inner radius a & outer radius b which carries a current *i* uniformly spread over its cross section. Show that the magnetic field B for points inside the body of the conductor (i.e. a < r < b) is given by, $B = \frac{\mu_0 i}{2\pi(b^2-a^2)}\frac{r^2-a^2}{r}$. Check this formula for the limiting case of a = 0.



(A) $\frac{\mu_0 i}{2\pi b^2}r$ (B) $\frac{\mu_0 i}{3\pi b^2}r$ (C) $\frac{\mu_0 i}{5\pi b^2}r$ (D) $\frac{\mu_0 i}{8\pi b^2}r$

20. The magnetic field B inside a long solenoid, carrying a current of 10 A, is 3.14×10^{-2} T. Find the number of turns per unit length of the solenoid.

(A) 1200 turns/m (B) 1500 turns/m (C) 2300 turns/m (D) 2500 turns/m

21. If a charged particle projected in a gravity-free room it does not deflect,

(A) electric field and magnetic field must be zero

(B) both electric field and magnetic field may be present

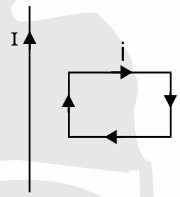
(C) electric field will be zero and magnetic field may be zero

(D) electric field may be zero and magnetic field must be zero

22. A particle having a charge of 2.0×10^{-8} C and a mass of 2.0×10^{-10} g is projected with a speed of 2.0×10^{3} m/s in a region having a uniform magnetic field (B = 0.1 T) perpendicular to it. Find the radius of the circle formed by the particle and also the frequency.

(A) 5 cm, $\frac{5}{\pi}\times10^{3}s^{-1}$ (B) 10 cm, $\frac{5}{\pi}\times10^{3}s^{-1}$ (C) 20 cm, $\frac{5}{\pi}\times10^{3}s^{-1}$ (D) 20 cm, $\frac{3}{\pi}\times10^{3}s^{-1}$

23. A rectangular loop carrying a current i is situated near a long straight wire such that the wire is parallel to one of the sides of the loop and the plane of the loop is same of the left wire. If a steady current I is established in the wire as shown in the (fig) the loop will -



(A) Rotate about an axis parallel to the wire (B) Move away from the wire

(C) Move towards the wire (D) Remain stationary.

24. Select the correct alternative(s):

Two thin long parallel wires separated by a distance 'b' are carrying a current 'i' ampere each. The magnitude of the force per unit length exerted by one wire on the other is

(A) $\frac{\mu_0 i^2}{b^2}$ (B) $\frac{\mu_0 i^2}{2\pi b}$ (C) $\frac{\mu_0 i}{2\pi b}$ (D) $\frac{\mu_0 i}{2\pi b^2}$

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ANSWER KEY

1.	(B)	**2.**	(A)	**3.**	(C)	**4.**	(B)	**5.**	(A)
6.	(C)	**7.**	(D)	**8.**	(A)	**9.**	(A)	**10.**	(D)
11.	(A)	**12.**	(B)	**13.**	(B)	**14.**	(B)	**15.**	(D)
16.	(D)	**17.**	(B)	**18.**	(AB)	**19.**	(A)	**20.**	(D)
21.	(BD)	**22.**	(C)	**23.**	(C)	**24.**	(B)		